Bihar (Patna Ganga floodplain)

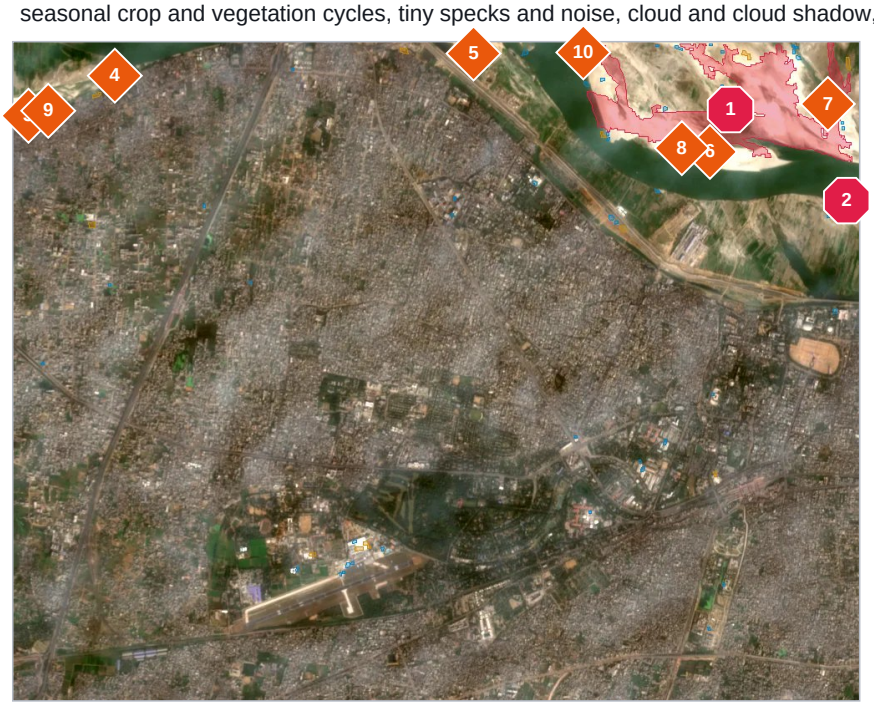
Patna (Danapur & Ganga riverfront fringe), Bihar | Before - Sentinel-2 - 2019-02-10 to After - Sentinel-2 - 2025-01-19

What we found

We found 79 real changes covering 202 hectares. The biggest and strongest (rank 1) is water receded: 169 ha at 25.6439, 85.1354.

By priority: 2 Critical, 8 High, 16 Medium, 53 Low.

The detector first flagged 2,049 candidate areas. The rest were removed as false alarms: change types this analysis does not track, seasonal crop and vegetation cycles, tiny specks and noise, cloud and cloud shadow, areas below the minimum size.



Numbers = rank of the 10 highest-priority changes. Outline colour = priority.

Changes by priority

Critical	2
High	8
Medium	16
Low	53

Totals

Area changed: 202 ha

Cropland affected: 20.28 ha (upper bound)

Built-up land affected: 3.61 ha

How to read this report

Rank	Every change has a number. Rank 1 is the highest priority: the largest and strongest change. Priority = size x strength of the change.
Priority colour and shape	Critical (octagon): score 20,000 or more (about 2 ha of strong change); High (diamond): score 5,000 or more; Medium (square): score 1,500 or more; Low (circle): score below 1,500 (small or weak).
Confidence	How sure the system is that the change is real, from 0 to 100 %. Changes marked 'needs a human check' sit between 50 % and 70 %.
What changed	New construction, vegetation cleared, vegetation growth, new water / flooding, water receded, or unexplained change. These come from rules on the satellite colours, not from knowing what is actually on the ground.
Raw detections	Everything the detector first flagged. Most are not real (clouds, crop seasons, shadows, specks), so only changes that pass the filters are listed here.

The 10 highest-priority changes

Start here. Each card says where the change is, how big it is, why it was flagged and what to do.

1

Water receded

Rank 1 of 79 | Critical priority

Size	169 ha (1,686,750 m2), about 236 football pitches
Confidence	98 %
Location	25.64385, 85.13539 (open in Google Maps)
What it is	Water that was there before has gone.
Why flagged	vegetation index rose (+0.35), water index fell (-0.46), SWIR/blue ratio (paved surfaces are flatter) rose (+1.00)
What to do	Check first. A large, strong change; send someone or review recent imagery now.

Before



After



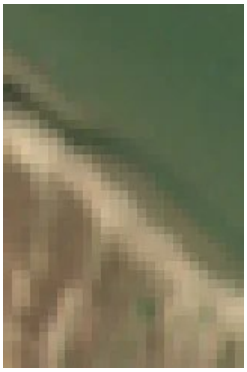
2

Water receded

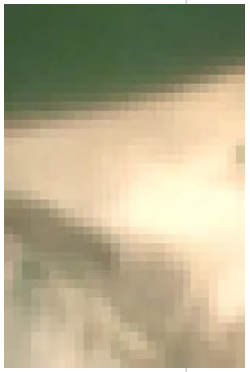
Rank 2 of 79 | Critical priority

Size	2.59 ha (25,900 m2), about 4 football pitches
Confidence	98 %
Location	25.63382, 85.14941 (open in Google Maps)
What it is	Water that was there before has gone.
Why flagged	built-up index rose (+0.21), vegetation index rose (+0.30), water index fell (-0.43), SWIR/blue ratio (paved surfaces are flatter) rose (+1.08)
What to do	Check first. A large, strong change; send someone or review recent imagery now.

Before



After



3

New construction

Rank 3 of 79 | High priority

Size	2.88 ha (28,850 m2), about 4 football pitches
Confidence	96 %
Location	25.64317, 85.05070 (open in Google Maps)
What it is	Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete.
Why flagged	vegetation index fell (-0.07), water index rose (+0.05), SWIR/blue ratio (paved surfaces are flatter) fell (-0.43)
What to do	Check soon. A substantial change that is probably real.

Before



After



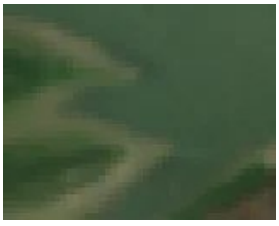
4

Water receded

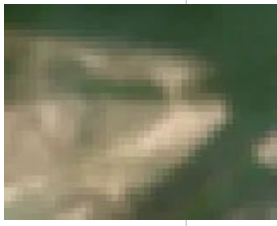
Rank 4 of 79 | High priority

Size	2.54 ha (25,350 m2), about 4 football pitches
Confidence	97 %
Location	25.64758, 85.06112 (open in Google Maps)
What it is	Water that was there before has gone.
Why flagged	built-up index rose (+0.22), vegetation index rose (+0.42), water index fell (-0.61), SWIR/blue ratio (paved surfaces are flatter) rose (+1.98)
What to do	Check soon. A substantial change that is probably real.

Before



After



5

New construction

Rank 5 of 79 | High priority

Size	2.08 ha (20,850 m2), about 3 football pitches
Confidence	97 %
Location	25.65003, 85.10439 (open in Google Maps)
What it is	Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete.
Why flagged	water index rose (+0.06), SWIR/blue ratio (paved surfaces are flatter) fell (-0.64)
What to do	Check soon. A substantial change that is probably real.

Before



After



6

New construction

Rank 6 of 79 | High priority

Size	2.36 ha (23,600 m2), about 3 football pitches
Confidence	96 %
Location	25.63920, 85.13291 (open in Google Maps)
What it is	Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete.
Why flagged	vegetation index fell (-0.12), water index rose (+0.14), SWIR/blue ratio (paved surfaces are flatter) fell (-0.89)
What to do	Check soon. A substantial change that is probably real.

Before



After



7

New construction

Rank 7 of 79 | High priority

Size	1.95 ha (19,450 m2), about 3 football pitches
Confidence	95 %
Location	25.64447, 85.14713 (open in Google Maps)
What it is	Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete.
Why flagged	SWIR/blue ratio (paved surfaces are flatter) fell (-0.37)
What to do	Check soon. A substantial change that is probably real.

Before



After



8

New construction

Rank 8 of 79 | High priority

Size	0.82 ha (8,250 m2)
Confidence	97 %
Location	25.63960, 85.12952 (open in Google Maps)
What it is	Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete.
Why flagged	vegetation index fell (-0.14), water index rose (+0.17), SWIR/blue ratio (paved surfaces are flatter) fell (-1.22)
What to do	Check soon. A substantial change that is probably real.

Before



After



9

New construction

Rank 9 of 79 | High priority

Size	0.93 ha (9,300 m2)
Confidence	96 %
Location	25.64372, 85.05309 (open in Google Maps)
What it is	Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete.
Why flagged	vegetation index fell (-0.09), water index rose (+0.06), SWIR/blue ratio (paved surfaces are flatter) fell (-0.34)
What to do	Check soon. A substantial change that is probably real.

Before



After



10

New construction

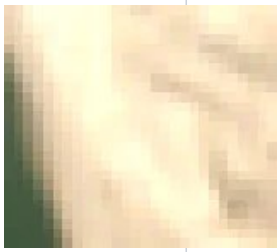
Rank 10 of 79 | High priority

Size	0.99 ha (9,900 m2)
Confidence	95 %
Location	25.65014, 85.11763 (open in Google Maps)
What it is	Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete.
Why flagged	vegetation index fell (-0.08), water index rose (+0.06), SWIR/blue ratio (paved surfaces are flatter) fell (-0.31)
What to do	Check soon. A substantial change that is probably real.

Before



After



All 79 changes found

Every change that survived the false-alarm filters, in priority order (rank 1 is the highest). A * after the confidence means it needs a human check. The CSV and GeoPackage downloads hold the same list with every attribute.

Rank	Priority	What changed	Size (ha)	Confidence	Latitude	Longitude
1	Critical	Water receded	169	98 %	25.64385	85.13539
2	Critical	Water receded	2.59	98 %	25.63382	85.14941
3	High	New construction	2.88	96 %	25.64317	85.05070
4	High	Water receded	2.54	97 %	25.64758	85.06112
5	High	New construction	2.08	97 %	25.65003	85.10439
6	High	New construction	2.36	96 %	25.63920	85.13291
7	High	New construction	1.95	95 %	25.64447	85.14713
8	High	New construction	0.82	97 %	25.63960	85.12952
9	High	New construction	0.93	96 %	25.64372	85.05309
10	High	New construction	0.99	95 %	25.65014	85.11763
11	Medium	New construction	0.94	94 %	25.64301	85.14659
12	Medium	New construction	0.64	97 %	25.63920	85.13492
13	Medium	New construction	0.55	97 %	25.64980	85.13733
14	Medium	New construction	0.52	98 %	25.63113	85.05843
15	Medium	New construction	0.45	99 %	25.59595	85.09154
16	Medium	New construction	0.54	97 %	25.63770	85.13410
17	Medium	New construction	0.53	96 %	25.65027	85.09608
18	Medium	Water receded	0.59	95 %	25.64884	85.14954
19	Medium	New construction	0.39	96 %	25.59486	85.08503
20	Medium	New construction	0.45	95 %	25.59548	85.09059
21	Medium	New construction	0.47	94 %	25.63064	85.12247
22	Medium	New construction	0.47	94 %	25.64454	85.14512
23	Medium	New construction	0.36	95 %	25.64531	85.05903
24	Medium	New construction	0.27	96 %	25.64101	85.12002
25	Medium	New construction	0.21	98 %	25.60365	85.13351
26	Medium	New construction	0.24	96 %	25.64906	85.10502
27	Low	Water receded	0.29	94 %	25.65005	85.11063
28	Low	New construction	0.26	95 %	25.64659	85.11797
29	Low	New construction	0.21	97 %	25.62159	85.14802
30	Low	New construction	0.28	94 %	25.65054	85.12909
31	Low	New construction	0.18	98 %	25.59324	85.08317
32	Low	New construction	0.25	95 %	25.59368	85.08930
33	Low	New construction	0.14	99 %	25.63990	85.13090
34	Low	New construction	0.22	96 %	25.59277	85.08858
35	Low	New construction	0.24	95 %	25.64354	85.04921
36	Low	New construction	0.28	93 %	25.65030	85.14313
37	Low	New construction	0.15	98 %	25.59623	85.09006
38	Low	New construction	0.24	94 %	25.60723	85.12748
39	Low	New construction	0.24	94 %	25.60425	85.12476
40	Low	New construction	0.30	92 %	25.63192	85.12098
41	Low	New construction	0.22	95 %	25.64384	85.12751
42	Low	Water receded	0.19	96 %	25.63483	85.12653
43	Low	New construction	0.23	94 %	25.64947	85.14351
44	Low	New construction	0.12	99 %	25.63872	85.13119
45	Low	New construction	0.18	95 %	25.63572	85.11184
46	Low	New construction	0.15	97 %	25.63917	85.12735
47	Low	New construction	0.17	95 %	25.64709	85.12000
48	Low	New construction	0.17	94 %	25.59541	85.09347

Rank	Priority	What changed	Size (ha)	Confidence	Latitude	Longitude
49	Low	New construction	0.12	98 %	25.64048	85.12066
50	Low	New construction	0.12	97 %	25.64111	85.12060
51	Low	New construction	0.12	97 %	25.63831	85.13035
52	Low	New construction	0.16	94 %	25.63138	85.12160
53	Low	Water receded	0.12	96 %	25.63221	85.10182
54	Low	New construction	0.14	94 %	25.64515	85.13823
55	Low	New construction	0.12	95 %	25.65010	85.13445
56	Low	New construction	0.15	92 %	25.64373	85.14612
57	Low	New construction	0.12	95 %	25.64175	85.14897
58	Low	New construction	0.10	96 %	25.59390	85.08986
59	Low	New construction	0.12	94 %	25.63402	85.10219
60	Low	New construction	0.12	94 %	25.63317	85.07193
61	Low	New construction	0.12	94 %	25.65070	85.11821
62	Low	New construction	0.09	97 %	25.60780	85.11674
63	Low	New construction	0.12	94 %	25.61580	85.05245
64	Low	New construction	0.08	97 %	25.65093	85.13305
65	Low	New construction	0.12	94 %	25.63189	85.14783
66	Low	New construction	0.12	93 %	25.64959	85.10603
67	Low	New construction	0.12	93 %	25.58666	85.13280
68	Low	New construction	0.09	96 %	25.64816	85.08264
69	Low	New construction	0.09	95 %	25.61244	85.13330
70	Low	New construction	0.09	94 %	25.62449	85.06062
71	Low	New construction	0.09	94 %	25.64233	85.14887
72	Low	New construction	0.09	94 %	25.64639	85.13436
73	Low	New construction	0.09	94 %	25.60510	85.12445
74	Low	New construction	0.09	93 %	25.59952	85.11846
75	Low	New construction	0.09	93 %	25.63202	85.14713
76	Low	Water receded	0.09	93 %	25.62480	85.07754
77	Low	New construction	0.09	93 %	25.59151	85.13452
78	Low	New construction	0.09	92 %	25.65075	85.14296
79	Low	New construction	0.08	94 %	25.65085	85.12657

Data sources

- Copernicus Sentinel-2 L2A (ESA), via the Element84 Earth Search COG bucket
- ESA WorldCover 2021 v200 (10 m), CC-BY 4.0

Limits - what this report cannot tell you

- Optical imagery cannot see through cloud; flagged areas with cloud are removed, not analysed.
- Change type comes from spectral index rules (no training); check critical changes on the ground.
- 10 m pixels cannot resolve single small buildings or people.
- Rivers, sandbars and lake edges move between dates and can be flagged as change; bare soil in dry regions can look like construction. Check flagged areas near water.
- Land cover is ESA WorldCover 2021, newer than the before image (2019): land already cleared or under construction can be mislabelled as cropland, so treat cropland as an upper bound
- Buildings and road segments need OpenStreetMap, which was not reachable when this result was built: shown as unknown, not zero
- This report points to where to look. Confirm important changes on the ground before acting.